

$\mathbb{Q}$ -vector space
 $\{1, \theta, \dots, \theta^{n-1}\}$
coords $a \in \mathbb{Q}^n$

matrix algebra
 $\rho: K \hookrightarrow M_n(\mathbb{Q})$
 $\rho(\theta) = C(m_\theta)$

eigenbasis M

$G = M^\top M$
trace form

Euclidean lattice
Minkowski σ
covol = $\sqrt{|d_K|}$